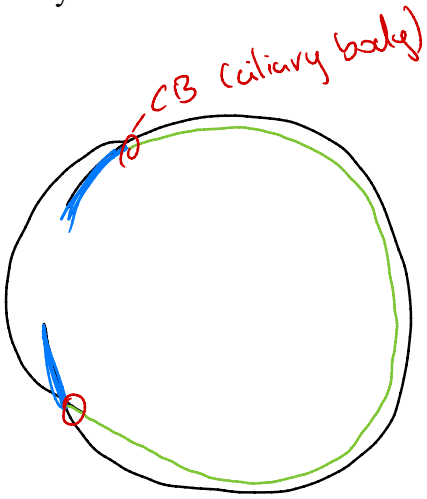
Uveal Tract (Uvea) (IRIS)

Images: 5.1, 5.27, 5.5, 5,17, 5.18, 5.21, 5.22, 5.26, 12.8, 5.29, 5.30, 12.15, 5.32, 5.35, 5.38

I. The uvea is the middle layer of the eye.

A. It is composed of 3 regions

iris - regulates light
CB - produces aqueous + accommodation
choroid - supply blood to retina



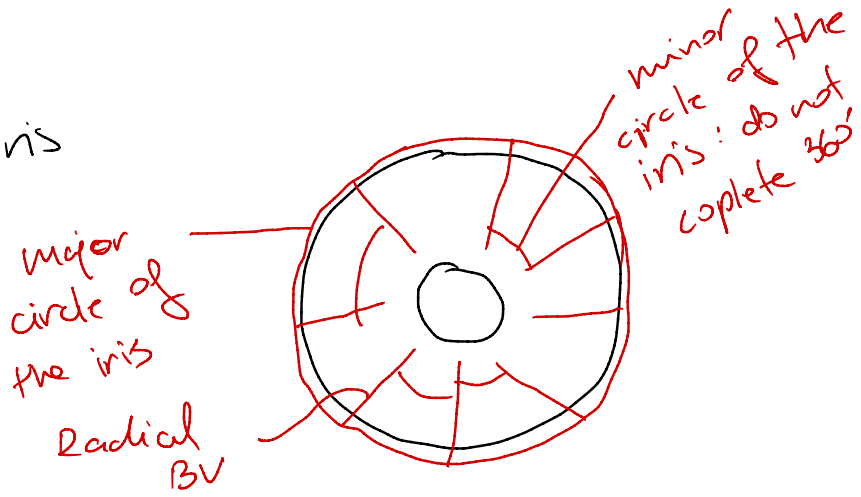
II. Iris

A. Iris characteristics

1. Iris root: peripheral termination of the iris around the edges 360°
   a. Joins the iris to the ciliary body
   b. Thinnest area of the iris
      This is were damage appears when comes to trauma so we have to look for edges of the iris
   c. Clinical correlate: iridodialysis
      (1) Iris root tears from the ciliary body

2. Blood vessels → BV

stroma of the iris



a. Branch from the major circle of the iris.

b. Radial blood vessels course from iris root to pupil margin

c. Minor circle of the iris

d. Iris capillaries are <u>not fenestrated</u> don't have hols in it. Very tight because we do not want to many

blood - aqueous barrier

4. Pupil:

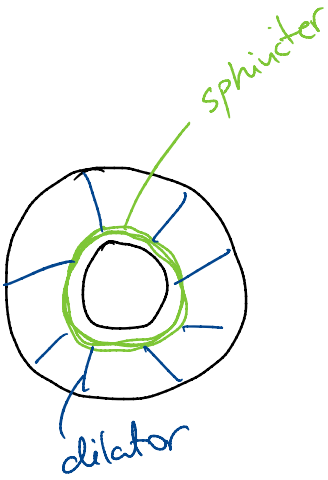
   a. Function: Regulate the light

   b. Size: 1 – 9 mm (depends on the light)

   c. Miotic: or mypsis; very small pupil (bright)

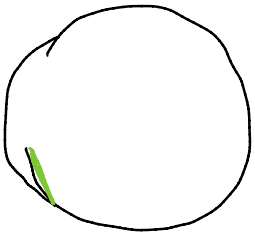
   d. Mydriatic: dilated pupil; dim ilumination
   mydriasis

   e. Sphincter muscle
   Smooth musle
   parasympathetic system
   Small pupil - miosis



   f. Dilator muscle:
   Smooth muscle
   Sympathetic system
   From iris root to
   big pupil - mydriasis

5. Posterior iris epithelium



   a. Heavily pigmented cells

   b. Clinical correlate: Pupillary ruff

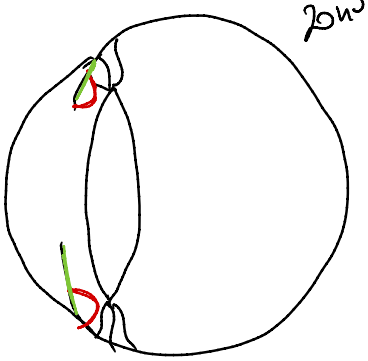
Darker than the rest of the iris

Postirior. Dark circle 

   c. Clinical correlate: Pigmentary dispersion syndrome

This can cause glaucoma.

      (1) Pigment from the posterior iris is dispersed into the anterior chamber

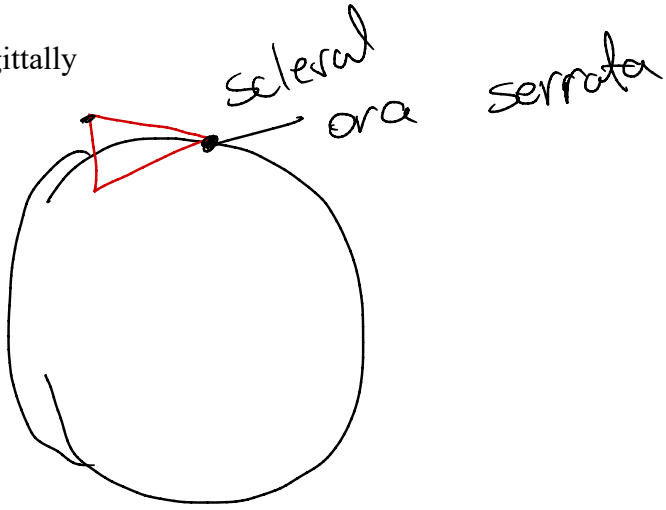


Zonules are not smooth so they rub the iris pigment and can cause the glaucoma

III. Ciliary body

A. Characteristics

1. Triangular shape sagittally



2. Ring-shaped structure from iris root to ora serrata

ant: processes
post: flat

3. Two parts

a. Pars plicata

ant
70-80 processes
+ valleys of Kuhnt (inbetween)
goes 360°

b. Pars plana

post. flatter
and that ends at ora serrata

(1) The zonular fibers

B. Ciliary muscle

1. Smooth muscle

2. Parasympathetic stimulation

contracts causes accommodation

3. Sympathetic stimulation has no clinically significant effect on the ciliary body.

C. Ciliary body blood vessels

1. Capillaries are fenestrated with wide lumens

D. Epithelium

1. Inner layer of cells are joined by zonula occludens.

2. Form part of the blood-aqueous barrier

3. Produces and secretes aqueous humor

Epithelium secrets

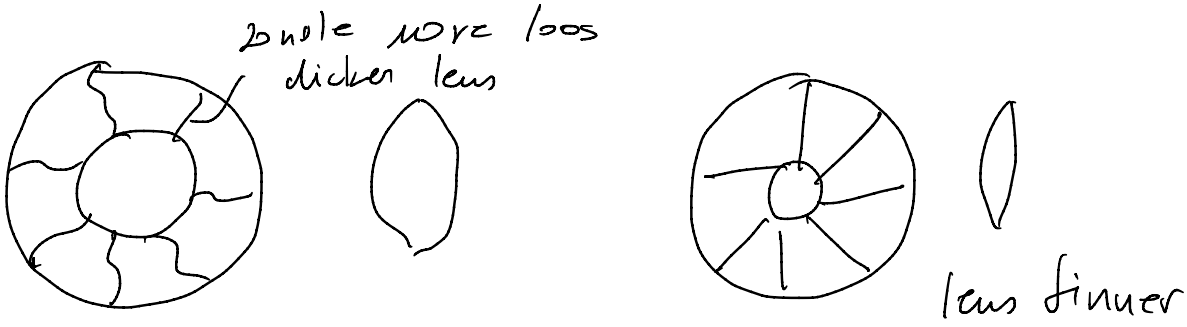
E. Ciliary body function

1. Produce and secrete aqueous

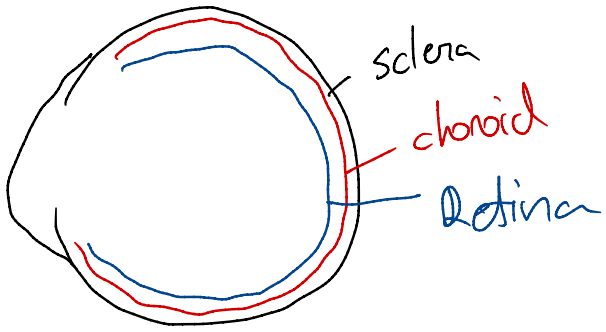
2. Part of the blood-aqueous barrier

a. Clinical correlate: Uveitis - inflamation in the eye
blood vessels in the iris.
leaking

3. Accommodation



IV. Choroid least of the 3



A. Choroidal arteries are branches of the short posterior ciliary arteries

short relatively large BV

1. The blood vessels get smaller as they branch and get closer to the retina.

2. Choriocapillaris

   a. Single layer of anastomosing, fenestrated capillaries

   b. Densest in the macular area: it is the sole blood supply to the macular area.

Macula only gets blood from choroid!

3. Choroidal veins join in vortex pattern (vortex veins) in each quadrant of the eye

B. Bruch's membrane (basal lamina) assentional part of our vision

a. Innermost layer of choroid

b. Tight adhesion between Bruch's membrane and RPE

(1) Clinical correlate: drusen and age-related macular degeneration ARMD

phospholipid build up
(age related macula degeneration)
if not treated → blindness